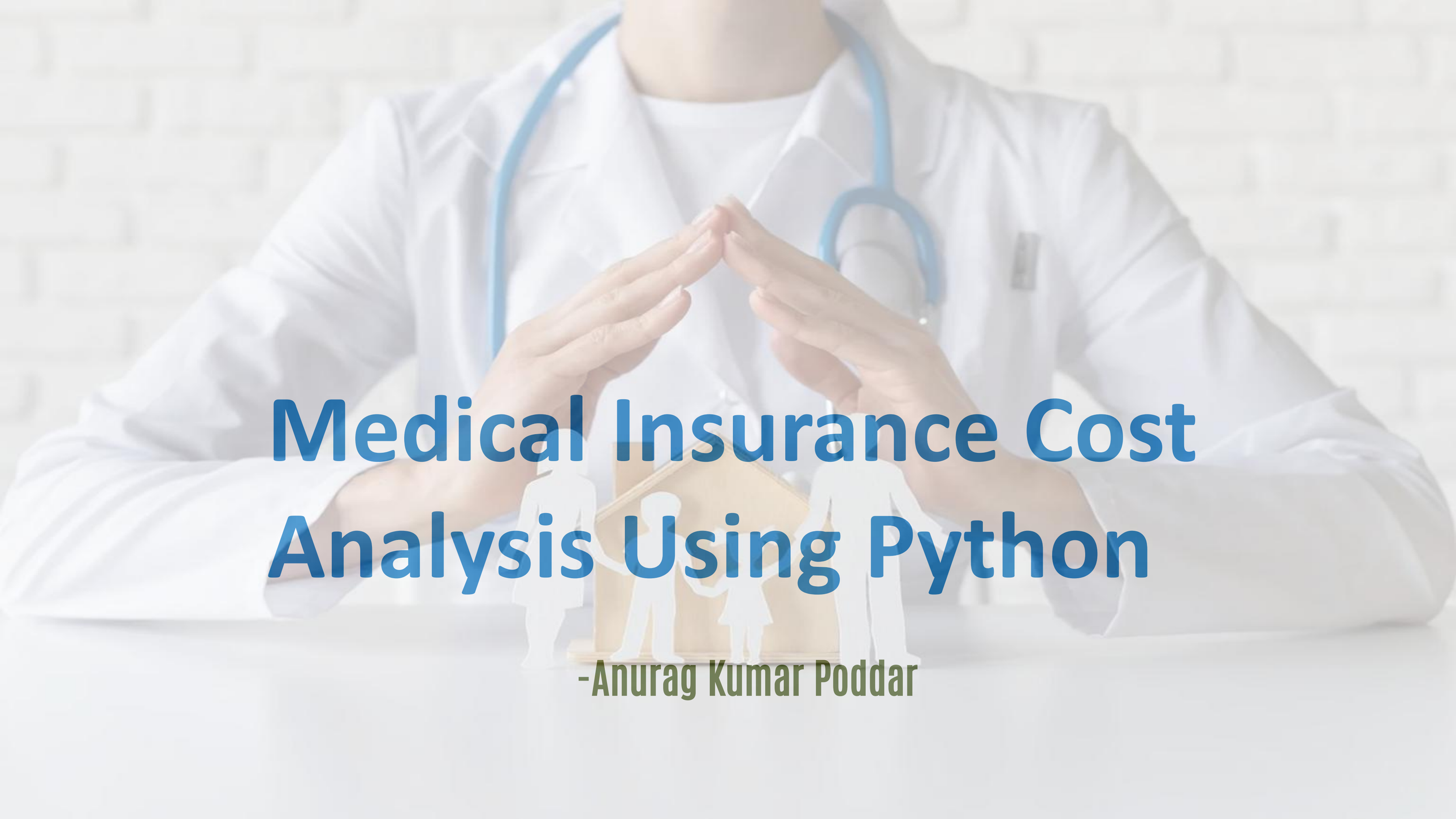


Medical Insurance Cost Analysis Using Python

-Anurag Kumar Poddar

Project Overview

- **This project analyzes healthcare insurance data using Exploratory Data Analysis (EDA).**
- **The objective is to understand how factors like age, BMI, smoking habits, and region affect medical insurance charges.**



Problem Statement

- **Healthcare costs vary significantly among individuals.**
- **This project aims to identify the key factors that influence insurance charges and understand their relationship with medical expenses.**

Dataset Information

Feature

- **age : Age of individual**
- **sex : Gender**
- **BMI : Body Mass Index**
- **children : Number of children**
- **smoker :Smoking status**
- **region : Residential region**
- **charges : Medical insurance cost**

Total Records: 1338

Total Features: 7

	A	B	C	D	E	F	G
1	age	sex	bmi	children	smoker	region	charges
2	19	female	27.9	0	yes	southwest	16884.924
3	18	male	33.77	1	no	southeast	1725.5523
4	28	male	33	3	no	southeast	4449.462
5	33	male	22.705	0	no	northwest	21984.47061
6	32	male	28.88	0	no	northwest	3866.8552
7	31	female	25.74	0	no	southeast	3756.6216
8	46	female	33.44	1	no	southeast	8240.5896
9	37	female	27.74	3	no	northwest	7281.5056
10	37	male	29.83	2	no	northeast	6406.4107
11	60	female	25.84	0	no	northwest	28923.13692
12	25	male	26.22	0	no	northeast	2721.3208
13	62	female	26.29	0	yes	southeast	27808.7251
14	23	male	34.4	0	no	southwest	1826.843
15	56	female	39.82	0	no	southeast	11090.7178
16	27	male	42.13	0	yes	southeast	39611.7577
17	19	male	24.6	1	no	southwest	1837.237
18	52	female	30.78	1	no	northeast	10797.3362
19	23	male	23.845	0	no	northeast	2395.17155
20	56	male	40.3	0	no	southwest	10602.385
21	30	male	35.3	0	yes	southwest	36837.467
22	60	female	36.005	0	no	northeast	13228.84695
23	30	female	32.4	1	no	southwest	4149.736
24	18	male	34.1	0	no	southeast	1137.011
25	34	female	31.92	1	yes	northeast	37701.8768



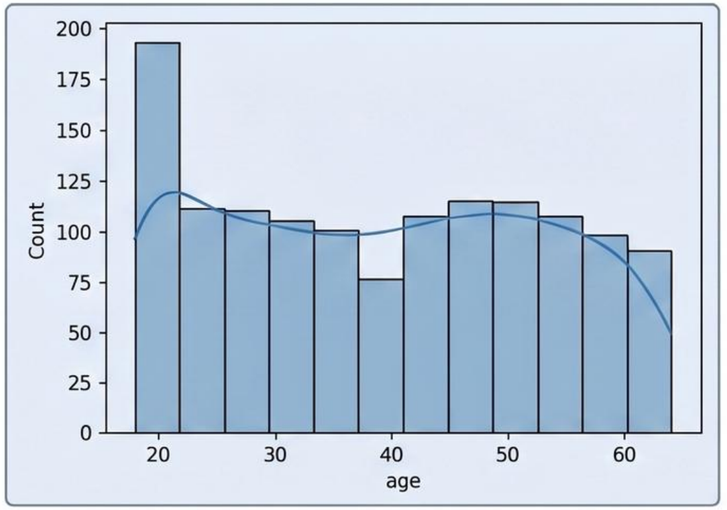


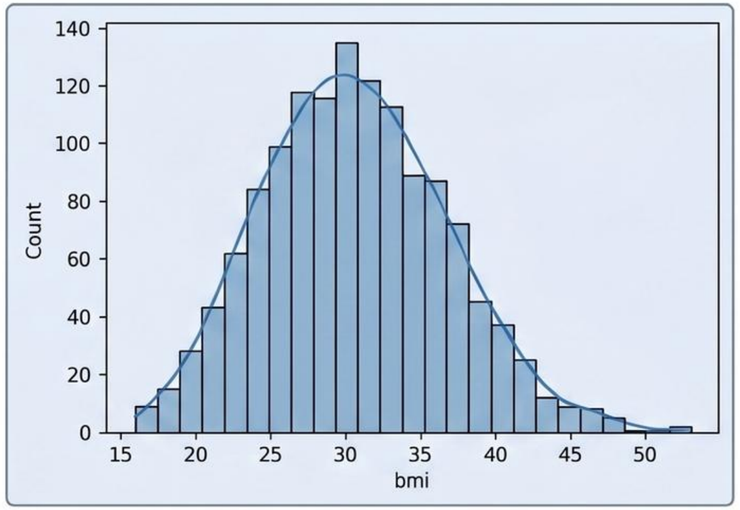
Data Cleaning

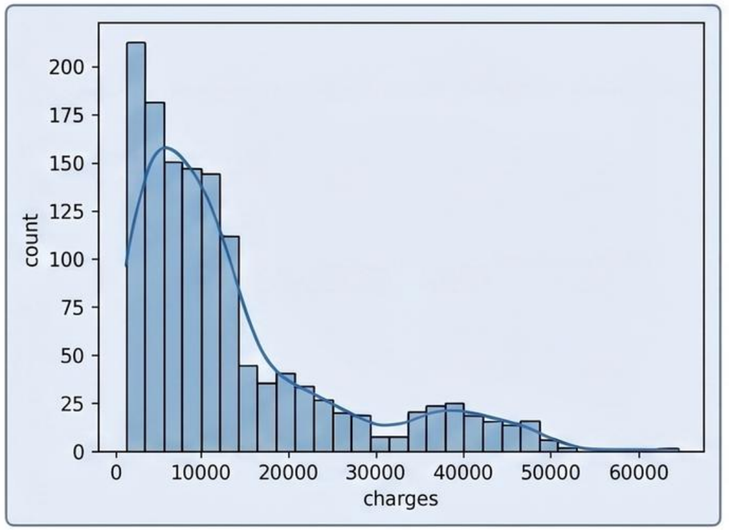
- Checked missing values
- Checked duplicate records
- Verified data types
- Dataset was clean and ready for analysis

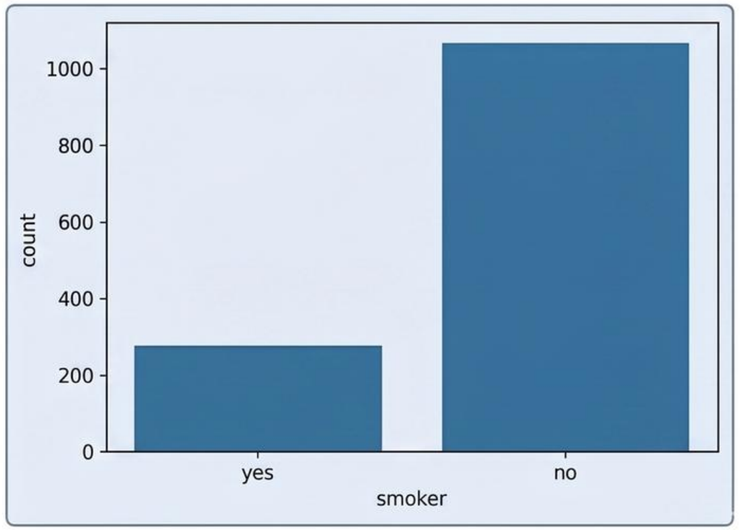
Univariate Analysis

- **Overall, the dataset shows a mix of balanced and skewed distributions.**
- **Numerical features like age and BMI are fairly well spread, while charges show a strong right skew with few high-cost cases.**
- **Categorical features such as sex and region are balanced, but smoker status is highly imbalanced, which is likely to be a key factor influencing insurance cost.**







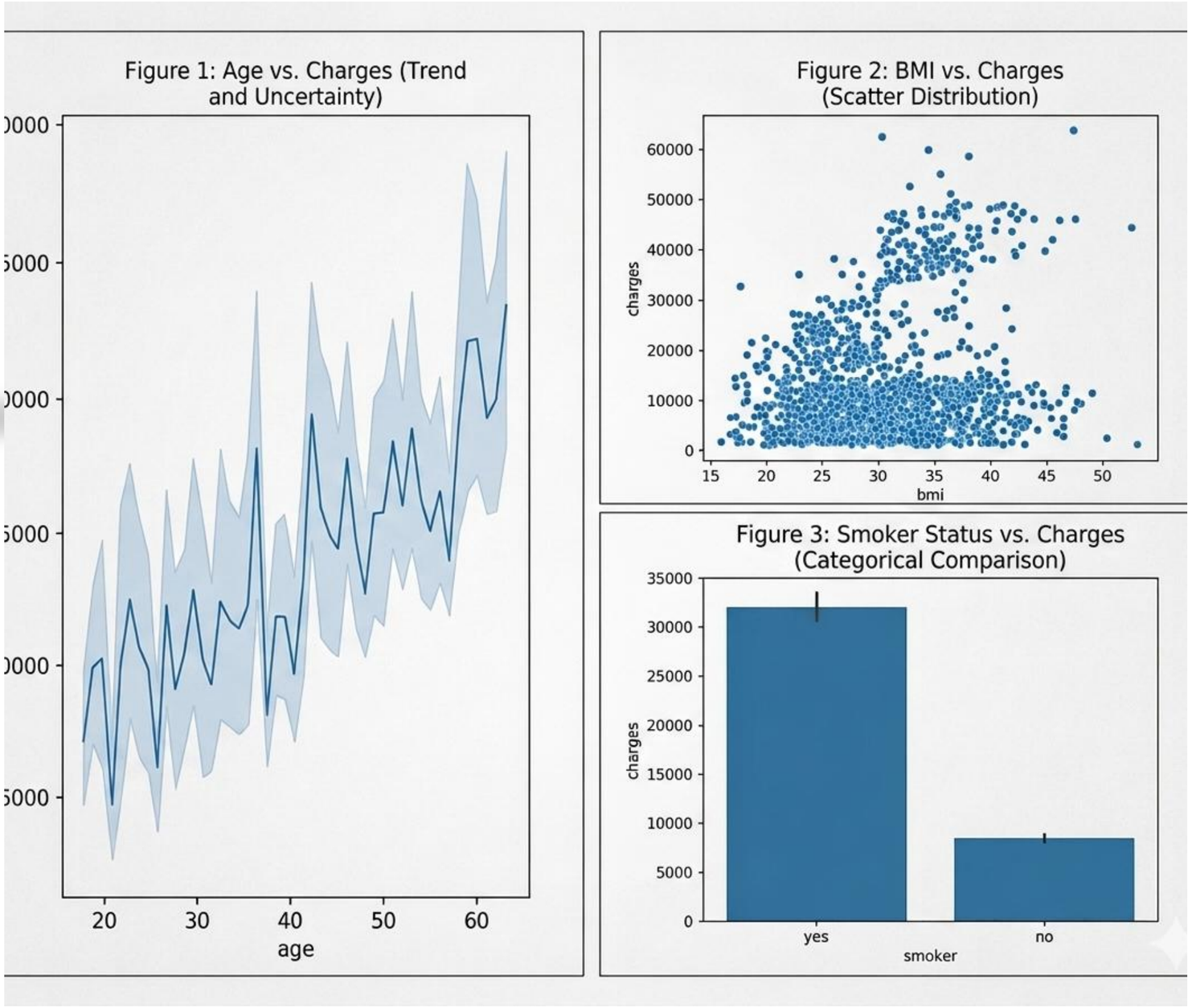


Bivariate Analysis

- **From the bivariate analysis, it is clear that insurance charges are not influenced equally by all features. The most important factor is smoking status, which shows a very strong impact—smokers have significantly higher charges compared to non-smokers.**

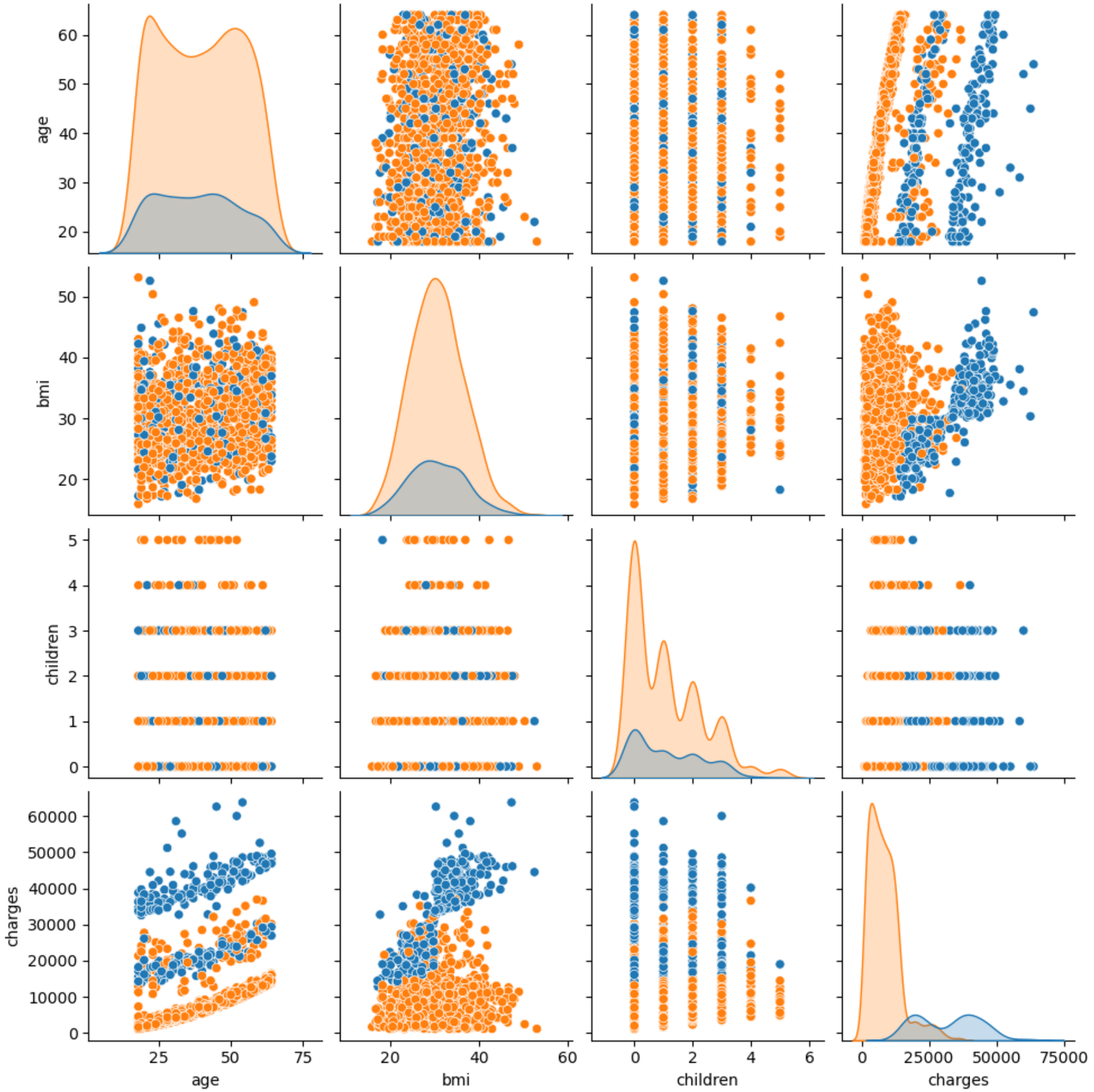
- **Other variables like age and BMI show a weak positive relationship with charges, indicating that costs increase slightly with higher age and body mass index, but the effect is not strong on its own.**

- **Overall, the analysis suggests that lifestyle-related factors (especially smoking) are the key drivers of medical insurance cost, while demographic factors play a secondary or minimal role.**



Multivariate Analysis

- The multivariate analysis reveals that insurance charges are influenced by the combined effect of multiple factors such as age, BMI, and smoking status. Among all variables, smoking emerged as the most significant factor associated with higher medical expenses.



Key Findings

Smoking is the most influential factor

BMI and age strongly affect charges

Non-smokers generally have lower expenses

Region has relatively smaller impact





Conclusion

- This project analysed medical insurance charges using Python and Exploratory Data Analysis (EDA).
- The results show that smoking status has the strongest observed association with insurance charges. Age and BMI have weaker positive relationships with charges, while the differences associated with sex, number of children, and region are less pronounced in the analysis.
- The project demonstrates how data cleaning, statistical summaries, and visualisations can help identify patterns in insurance data. Further statistical testing and machine learning can be used to develop a more detailed understanding of these relationships.

Thank you

